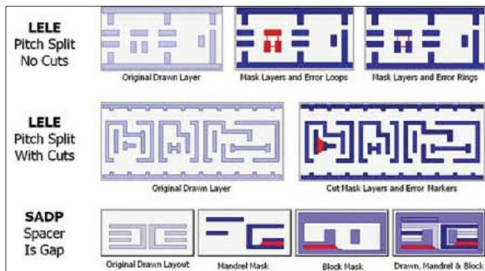


Handling 20nm Design Challenges

While the world is working on 22nm manufacturing process, plans are underway for 20 nm and beyond. At 20 nm, there are various issues to deal with, such as complex double-patterning lithography requirements, in-design physical verification throughout the flow, accurate higher levels of extraction and timing analysis to allow for manufacturing variability, and so on. How is the global semiconductor industry handling these and a host of other issues? Find out



Double-patterning lithography (Courtesy: www.semiwiki.com)

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1 Complex double-patterning lithography

Complex double-patterning lithography in 20nm process requires rule-aware placement and routing in order to ensure the ability to colour mask correctly and efficiently.

Anil Kalra, senior group manager, Design Enablement & Services, STMicroelectronics, says, "With nanotechnology continuing to allow shrinking of semiconductor devices, VLSI design is becoming more and more exciting for designers. It is an amazing industry where people are continuing to bring innovative solutions for seemingly blocking challenges being faced along the way. Crossing the available litho limits with multi masks for a layer (dual and triple patterning) is becoming a reality—although at a price and increased complexity to be managed both by the designers and the manufacturers—possible only by close col-

laboration with EDA industry."

Jaswinder Ahuja, corporate VP and MD, Cadence Design Systems (I), president, VLSI Society of India, and former ISA chairman, adds: "One of the most difficult (and discussed) design challenges at 20 nm deals with the need for double patterning to overcome the limits of existing lithography equipment. It has such a pervasive impact on the silicon that every step in the design flow must anticipate and prevent potential problems. Standard cell development, placement, clocking, routing, extraction, timing and power analysis, and design-rule checking need to be 'double-patterning aware.' Therefore a unified, double-patterning-aware design flow across custom and digital domains becomes imperative at 20nm."

The first thing affected by double patterning is cell and library generation. Designers need to make sure that the silicon IP is compliant with double-patterning layout rules. It is also critical to take double patterning

into account during placement.

Cadence has developed a technology that automates coloured placement, resulting in a less congested design that helps meet timing and power requirements. However, the most important aspect is routing. Double patterning has to be integrated in the routing solution. It cannot be an afterthought where you complete routing and then run decomposition. It has to be done correct-by-construction and that is Cadence's approach.

Cadence carries double-patterning intent forward from cell and IP generation to double-pattern-aware routing, and finally to signoff physical verification. This provides faster convergence because the intent is carried forward throughout the flow.

Andy Biddle, product marketing manager, Implementation Group, Synopsys, informs, "We have been working with the leading foundries and early adopter customers to develop an effective solution to this problem. We have made a number of enhancements to IC Compiler in order to ensure a fully decomposable design with minimal die overhead."

To ensure minimal die area, there is much more than just making space between cells and tracks. Work starts with library preparation stage by analysing the library and identifying cells that are already DPT clean (cells that can be abutted to other cells). If the colour information is provided in the library, this will be used by the placer.

DPT clean means that the placer must ensure adjacent cells meet minimal space requirements if the boundaries are of the same colour or,